

The semi-quasi-homogeneous arc: a synthesis (germbij class (c), three tides to the consult endpoint)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-10

Abstract

Three tides (PRs #113, #114, #116) formalized the recoverable fragments of the germbij note’s constructive class (c) — the semi-quasi-homogeneous isolated minima of §7’s one-point anchoring — and stopped at a boundary drawn in advance by a scoping consult. The results: for a quasi-homogeneous principal part P , the normalized monomial moments recover every moment ratio *exactly*, $t^{\langle q, \alpha \rangle} \langle x^\alpha \rangle_t = M_\alpha / M_0$ for all $t > 0$ and every multi-index including odd ones (**QhMomentRecovery**); for two Gibbs families over any common reference density whose corrections differ at rate h^ρ toward Q , the normalized moments separate at exactly that rate with coefficient $-\text{Cov}_P(A, Q)$ (**GradeComparison**); and matching moment rates therefore force each weighted correction grade to agree, by covariance injectivity (**GradeRecovery**). Together: principal-model moment ratios are recoverable, and conditional on a common recovered principal part, each finite weighted correction grade is recoverable from normalized moment rates — the note’s covariance argument for class (c), made checkable, minus the coarea layer.

1 The consult’s two findings

The scoping consult reshaped the work before it began. First, the note’s recovery of the moments M_α proceeds through a linear-independence argument over partial derivatives of delta distributions, with care at exponent collisions; the consult observed that testing with the monomial observable itself dissolves the problem — collisions between different multi-indices never enter a single observable’s leading coefficient — so the formalisation needs no distribution theory at all, and the recovery is an exact identity rather than an asymptotic. Second, it drew the endpoint: the one-grade comparison is the reusable core; the coarea/Gelfand–Leray identification of the principal part from its moments awaits real geometric measure theory in Mathlib, and the full weighted-analytic induction (truncations, anisotropic remainders, germ summation) is a programme, not a tide. Both findings held exactly.

2 What each tide contributed

- **QhMomentRecovery** (#113): `mvMonomial` and its homogeneity along the quasi-homogeneous dilation; the exact unnormalized and normalized monomial moment laws (instantiating the degenerate-perimeter arc’s abstract **ScalesMeasure** laws at

the diagonal-volume instance); the headline identity and the single-temperature comparison form. One build round.

- **GradeComparison** (#114): the non-Gaussian covariance API (`expectationUnder`, `covarianceUnder`); the pointwise engine $(e^{-W} - 1)/h^p \rightarrow -Q$ via the second-order bound, never dividing by the correction; the unnormalized dominated-convergence limit; the normalized limit through the quotient-rate identity. Numerically verified first.
- **GradeRecovery** (#116): covariance bilinearity over monomial combinations; the L^2 variance rigidity (almost-everywhere-to-everywhere by continuity against the open-positive volume, constant killed at the origin); and the stage-D composition by uniqueness of limits, whose interface hypothesis is stage B’s conclusion — callers discharge the majorant package, this file stays free of domination bookkeeping.

3 Relation to the note

The note’s paragraph on semi-quasi-homogeneous recovery (§7, constructive classes, item (c)) now carries four margin markers: the exponent-affinity/moment-ratio sentence points at `qh_momentRatio_recovery` and the comparison form; the covariance-and-variance sentence (“their difference Q has $\text{Var}_P[Q] = 0$, so Q is constant, and positive weighted degree forces $Q = 0$ ”) points at `tendsto_normalized_difference_div_pow` and `grade_eq_of_normalized_rates`. The note’s further steps — identifying P itself from the moment measure on $\{P = 1\}$ (coarea), and summing the recovered weighted jet to the analytic germ — are exactly the two items the consult placed beyond the endpoint.

4 What remains open, and why

Piece (ii), the coarea/Gelfand–Leray layer: a Mathlib prerequisite, not a proof-engineering gap. The full weighted-analytic induction: a programme of weighted truncation infrastructure that the commissioning direction ranked below its cost. Concrete anisotropic instantiations of stages B–D for named weighted families: available on demand, the abstract theorems were shaped for it. The class closes here by design; per the arc’s tide logs, it should not be extended without new direction.